Spatial Truth: Location Service for Physical AI

APPENDICES FAQ (DEEP DIVES)

APPENDIX D: Technical Overview

Executive Summary For Centimeter-level accuracy, Spatial Truth uses AI-driven signal reconstruction and tightly integrated 5G–NTN positioning. This appendix outlines how these components work together inside the network core.

Q D1: Architecturally, what is the core shift?

A: Transition from Device-Centric to Network-Centric. Current high-accuracy positioning relies on "Hardware Clutter" (LiDAR/RTK) installed on every asset, creating massive CAPEX and technical debt. Spatial Truth utilizes the Network-as-a-Sensor. The platform processes raw signal data at the edge (the 5G Core) and delivers centimeter accuracy to the device via a unified abstraction layer. This effectively upgrades the positioning capabilities of the network without requiring physical modifications to the vehicle.

Q D2: How is "location uncertainty" resolved in the Urban cities?

A: GPS fails in dense cities because signals bounce off glass and steel. Instead of attempting to filter these reflections as noise, the Spatial Truth AI treats them as a unique environmental signature. Super-Resolution Neural Networks within the LMF (Location Management Function) calculate the direct path among the interference.

The Result: Physics-based interference is transformed into a positioning strength, achieving sub-meter accuracy.

Q D3: How is a "Single API" possible across network vendors (Ericsson, Huawei, Nokia)?

A: Spatial Truth functions as a Northbound Middleware.

- The Interface: While the "Southbound" integration requires specific adapters for different Network Function (NF) vendors (Ericsson, Huawei, etc.), Spatial Truth abstracts this complexity.
- The Benefit: Enterprise customers (e.g., Amazon, Toyota) interact with a single, standardized API. The platform handles underlying handshakes between the Cisco IoT Control Center and various carrier-side LMF implementations, ensuring a consistent data format regardless of the local hardware vendor.

Q D4: How are connectivity rural "Dead Zones" managed?

A: Spatial Truth ensures a continuous **"source of truth"** by blending three distinct layers into a single coordinate stream:

- Urban: 5G + Wi-Fi RTT (Passive ranging).
- Highway: 5G + V2X (Safety bands).
- Rural/Remote: NTN (Satellite bridge via Skylo).